ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Linear Response Theory and The Kubo Formula

I. The Bridge Between Macroscopic Transport and Microscopic Dynamics

II. Density Matrix and Correlation Functions

III. The Kubo Formula

IV. Green's Function Implementation

I. The Bridge Between Macroscopic Transport and Microscopic Dynamics

In classical Drude theory, conductivity, σ, is treated as a phenomenological constant derived from scattering times

Ohm's Law: $J = \sigma \cdot E$

In quantum systems (like Graphene), electrons behave as waves, necessitating a framework that treats transport as a quantum transition process

Macroscopic transport is the statistical manifestation of underlying microscopic quantum dynamics

I. The Bridge Between Macroscopic Transport and Microscopic Dynamics

Consider a system defined by H_0 and apply a weak external electric field $E(t)$

This field is represented via a vector potential, A(t)

$$E(t) = -\partial_t A(t)$$

The total Hamiltonian becomes $H(t) = H_0 + V(t)$

Using minimal coupling $p \rightarrow p + eA$

Interaction term: $$V(t) = -\int J(r) \cdot A(r,t)\, d^3r$$

The external electric field acts as a time-dependent perturbation to the equilibrium Hamiltonian

I. The Bridge Between Macroscopic Transport and Microscopic Dynamics

Density matrix: $\rho(t) = \rho_0 + \rho_1(t)$

Density operator in the equilibrium state — **The first-order correction based on the perturbation**

Liouville Equation: $\frac{\partial \rho}{\partial t} = -[\rho, H] \longrightarrow i\hbar \frac{\partial \rho}{\partial t} = [H_0 + V, \rho_0 + \rho_1]$

$$i\hbar\left(\frac{\partial \rho_0}{\partial t} + \frac{\partial \rho_1}{\partial t}\right) = [H_0 + V, \rho_0 + \rho_1] \approx [H_0, \rho_0] + [H_0, \rho_1] + [V, \rho_0] + [V, \rho_1]$$

$[H_0, \rho_0] = 0$ — Commute with each other

$[V, \rho_1]$ — 2nd order perturbation—ignore

$$i\hbar \frac{\partial \rho_1}{\partial t} \approx [H_0, \rho_1] + [V, \rho_0]$$

Define the interaction-picture density matrix $\tilde{\rho}_1(t)$ **as:** $\tilde{\rho}_1(t) = e^{iH_0 t/\hbar} \rho_1(t) e^{-iH_0 t/\hbar}$

Define the interaction-picture perturbation $\tilde{V}(t)$ **as:** $\tilde{V}(t) = e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar}$

I. The Bridge Between Macroscopic Transport and Microscopic Dynamics

$$\frac{d\tilde{\rho}_1}{dt} = \left(\frac{iH_0}{\hbar}\right) e^{iH_0t/\hbar}\rho_1(t)e^{-iH_0t/\hbar} + e^{iH_0t/\hbar}\dot{\rho}_1(t)e^{-iH_0t/\hbar} + e^{iH_0t/\hbar}\rho_1(t)\left(-\frac{iH_0}{\hbar}\right)e^{-iH_0t/\hbar}$$

$$\frac{d\tilde{\rho}_1}{dt} = \frac{i}{\hbar}[H_0, \tilde{\rho}_1] + e^{iH_0t/\hbar}\left(\frac{1}{i\hbar}[H_0, \rho_1] + \frac{1}{i\hbar}[V, \rho_0]\right)e^{-iH_0t/\hbar}$$

$$-\frac{i}{\hbar}[H_0, \tilde{\rho}_1]$$

The linear response is mathematically captured by the first-order perturbation of the density matrix relative to the equilibrium state

$$\frac{d\tilde{\rho}_1}{dt} = \frac{1}{i\hbar}[\tilde{V}, \tilde{\rho}_0] \dashrightarrow \tilde{\rho}_1 = \frac{1}{i\hbar}\int_{-\infty}^{t}[\tilde{V}(t'), \tilde{\rho}_0]dt'$$

$$\rho_1 = -\frac{i}{\hbar}\int_{-\infty}^{t} e^{iH_0t/\hbar}[\tilde{V}(t'), \tilde{\rho}_0]e^{-iH_0t/\hbar}dt'$$

$$\rho_1 = -\frac{i}{\hbar}\int_{-\infty}^{t} e^{-iH_0(t-t')/\hbar}[V(t'), \rho_0]e^{iH_0(t-t')/\hbar}dt'$$

II. Density Matrix and Correlation Functions

The evolution of ρ_1 is driven solely by the commutator $[V(t'), \rho_0]$. This implies that if the perturbation, V, commutes with the equilibrium density matrix ρ_0, the system shows no linear response to that specific driving force

The fundamental mechanism of response is the coupling between the perturbation operator and the equilibrium state

The expectation value of the current at time *t* is $\langle J_\mu(t)\rangle = Tr[\rho_1(t), J_\mu]$

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} Tr\{[V(t'), \rho_0], J_\mu(t-t')\}dt'$$

The current operator in the Heisenberg picture

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} Tr\{[V(t')\rho_0 - \rho_0 V(t')], J_\mu(t-t')\}dt'$$

$$Tr([A,B],C) = Tr([B,C],A)$$

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} Tr\{[\rho_0 J_\mu(t-t')V(t') - \rho_0 V(t')J_\mu(t-t')], \}dt'$$

The macroscopic current response is determined by the trace of the commutator between the current operator and the perturbation operator

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} Tr\{\rho_0[J_\mu(t-t')V(t')]\}dt'$$

II. Density Matrix and Correlation Functions

$$V(t) = -\int J(r)\cdot A(r,t)\,d^3r$$

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} Tr\left\{\rho_0,\left[J_\mu(r,t),-\int J(r',t')\cdot A(r',t')\,d^3r'\right]\right\}dt'$$

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} dt'\int d^3r'\,Tr\{\rho_0,[J_\mu(r,t),J_\nu(r',t')]\}\cdot A_\nu(r',t')$$

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} dt'\int d^3r'\,\Phi_{\mu\nu}(r,t;r',t')A_\nu(r',t')$$

This response operator relates the stimuli, injected current at position r' and time, t' from the environment, with the response from the system at position r and time t

Correlation Function $\Phi_{\mu\nu}(r,t;r',t') = \frac{i}{\hbar}\Theta(t-t')Tr\{\rho_0,[J_\mu(r,t),J_\nu(r',t')]\} = \frac{i}{\hbar}\langle[J_\mu(r,t),J_\nu(r',t')]\rangle_0$

Partition function $\rho_0 = \frac{e^{-\beta(H_0-\mu N)}}{Z}$; **where,** $\beta = \frac{1}{k_BT}$ **and** μ **is the chemical potential**

II. Density Matrix and Correlation Functions

$$\langle J_\mu(t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} dt' \int d^3r'\, \Phi_{\mu\nu}(r,t;r',t')A_\nu(r',t')$$

Relate the vector potential *A* back to the electric field *E*. By performing a Fourier transform into the frequency domain (ω), time-convolution is converted into a product of terms, leading directly to the definition of $\sigma_{\mu\nu}(\omega)$

Conductivity is defined as the frequency-domain response of the current to an applied electric field

III. The Kubo Formula

The Kubo formula is given by: $\sigma_{\mu\nu}(\omega) = \frac{1}{V}\int_0^\infty dt e^{i\omega t}\int_0^\beta d\lambda \langle [J_\mu(t), J_\nu(-i\hbar\lambda)]\rangle_0$

Separates the temporal evolution of the current operator and the thermal averaging over the equilibrium state

This Kubo formula expresses transport coefficients solely in terms of equilibrium state correlation functions

To ensure the integral converges as t to ∞, replace ω with $\omega + i\eta$. This η represents an infinitesimal adiabatic switching of the field, reflecting the fact that the system has finite memory of the perturbation

The introduction of η is essential to satisfy the causality requirements of the linear response

$$\sigma_{\mu\nu}(\omega + i\eta) = \frac{1}{V}\int_0^\infty dt e^{i\omega t}\left[e^{-\eta t}\Phi_{\mu\nu}(t)\right]$$

The real part of the conductivity $Re[\sigma(\omega)]$ is proportional to the spectral density of equilibrium current fluctuations. This links external energy dissipation (Joule heating) to internal microscopic current fluctuations (Brownian motion of electrons)

The dissipation of energy in an external field is fundamentally tied to the intrinsic quantum noise of the equilibrium system

III. The Kubo Formula

The power dissipation density is $P = J \cdot E$. Averaged over time, $P = \frac{1}{2} Re[\sigma(\omega)]|E(\omega)|^2$, the system absorbs energy, converting electrical energy into heat through scattering processes

The real part of the Kubo formula directly quantifies the rate of energy transfer from the external field to the quantum system: $J(t) = Re\left[\sigma(\omega) E_0 e^{-i\omega t}\right]$

Because the response must be causal (no current before the field is applied), the real and imaginary parts of $\sigma(\omega)$ are Hilbert transforms of each other. $Re[\sigma(\omega)] = \frac{1}{\pi} P \int \frac{Im[\sigma(\omega')]}{\omega' - \omega} d\omega'$

Causality forces the complex components of the conductivity to be inextricably linked via the Kramers-Kronig relations

$$\frac{i}{\hbar}\left\langle \left[J_\mu(r,t), J_\nu(r',t')\right]\right\rangle_0 = \frac{i}{\hbar}\Theta(t-t')Tr\left\{\rho_0, \left[J_\mu(r,t), J_\nu(r',t')\right]\right\}$$

Induce the Hamiltonian, H, with eigen vector $|n\rangle$

$$\sum_n |n\rangle\langle n| = 1$$

$$J_\nu(-i\hbar\lambda) = e^{\lambda H} J_\nu e^{-\lambda H}$$

$$\left\langle \left[J_\mu(r,t), J_\nu(r',t')\right]\right\rangle_0 = \sum_{n,m} \frac{e^{-\beta E_n}}{Z} \langle n|J_\nu|m\rangle\langle m|J_\mu|n\rangle e^{i(E_n - E_m)t/\hbar}$$

III. The Kubo Formula

$$\sigma_{\mu\nu}(\omega) = \frac{1}{V}\int_0^{\infty} dt e^{i\omega t}\int_0^{\beta} d\lambda \left\langle \left[J_\mu(t), J_\nu(-i\hbar\lambda)\right]\right\rangle_0$$

$$\int_0^{\infty} dt e^{i\left(\omega+\frac{E_n-E_m}{\hbar}\right)t} \rightarrow \pi\delta(\hbar\omega + E_n - E_m)$$

$$\left\langle \left[J_\mu(r,t), J_\nu(r',t')\right]\right\rangle_0 = \sum_{n,m}\frac{e^{-\beta E_n}}{Z}\langle n|J_\nu|m\rangle\langle m|J_\mu|n\rangle e^{i(E_n-E_m)t/\hbar}$$

$$J = ev$$

$$\left\langle \left[J_\mu(r,t), J_\nu(r',t')\right]\right\rangle_0 = e^2\sum_{n,m}\frac{e^{-\beta E_n}}{Z}\langle n|v|m\rangle\langle m|v|n\rangle e^{i(E_n-E_m)t/\hbar}$$

$$\frac{e^{-\beta E_n} - e^{-\beta E_m}}{Z} \approx f(E_n) - f(E_m)$$

Fermi-Dirac distribution function

As $\omega \rightarrow 0$, $f(E_n) - f(E_m) \approx (E_m - E_n)\left(-\frac{df}{dE}\right)$

$$\left\langle \left[J_\mu(r,t), J_\nu(r',t')\right]\right\rangle_0 = e^2\sum_{n,m}\langle n|v|m\rangle\langle m|v|n\rangle(E_m - E_n)\left(-\frac{df}{dE}\right)$$

$$\delta(E-H) = \sum_n |n\rangle(E-E_n)\langle n|$$

$$\left\langle \left[J_\mu(r,t), J_\nu(r',t')\right]\right\rangle_0 = e^2\sum_{n,m}\langle n|v_\mu|m\rangle\langle m|v_\nu|n\rangle\delta(E-E_n)\delta(E-E_m)\left(-\frac{df}{dE}\right)$$

$$\sigma_{\mu\nu}(\omega) = \frac{\pi e^2\hbar}{V}\int dE\left(-\frac{df}{dE}\right)Tr\left[v_\mu\delta(E-E_n)v_\nu\delta(E-E_m)\right]$$

IV. Green's Function Implementation

To compute the formula numerically, move from the time domain to the energy domain

Define the Retarded Green function as $G^R(E) = (E - H + i\eta)^{-1}$ The parameter η provides the broadening necessary to resolve the continuous spectrum from discrete states

The Advanced Green function is defined as $G^A(E) = (E - H - i\eta)^{-1} = [G^R(E)]^\dagger$

As $\eta \to 0$

$$G^R(E) - G^A(E) = -2\pi i \delta(E - H)$$

$$\sigma_{\mu\nu}(\omega) = \frac{\pi e^2 \hbar}{V} \int dE \left(-\frac{df}{dE}\right) Tr\left[v_\mu \delta(E - E_n) v_\nu \delta(E - E_m)\right]$$

Kubo-Greenwood Formula $$\sigma_{\mu\nu}(\omega) = \frac{\pi e^2 \hbar}{V} \int dE \left(-\frac{df}{dE}\right) Tr\left\{v_\mu \left[G_n^R(E) - G_n^A(E)\right] v_\nu \left[G_m^R(E) - G_m^A(E)\right]\right\}$$

A computationally efficient way to calculate conductivity using Hamiltonian matrix elements

In practice, one evaluates $Tr\left\{v_\mu \left[G_n^R(E) - G_n^A(E)\right] v_\nu \left[G_m^R(E) - G_m^A(E)\right]\right\}$ by diagonalizing H or by using recursive Green's function methods. For large systems, one utilizes sparse matrix multiplication to avoid the O(N^3) cost of full diagonalization

IV. Green's Function Implementation

♦ **Kubo theory inherently includes these effects. Kubo is necessary for systems where the scattering rate is comparable to the energy level spacing (*e.g.*, highly disordered systems)**

Kubo theory provides a more rigorous quantum mechanical description than the semi-classical Boltzmann approach

♦ **In reality, η represents the scattering rate due to extrinsic impurities or lattice defects. Without η, the conductivity would diverge at the Dirac point. Thus, η is not just a numerical trick, but a physical parameter describing system lifetime**

The broadening parameter η physically corresponds to the inverse lifetime of the electronic states due to scattering

Summary

Component	Mathematical Role	Physical Meaning
ρ_0	Equilibrium State	System background
J	Current Operator	Transport channel
G^R	Resolvent Operator	Propagator of excitations
η	Broadening	Scattering/Decay rate



That's all for this session

Thank you for your attention and great participation